

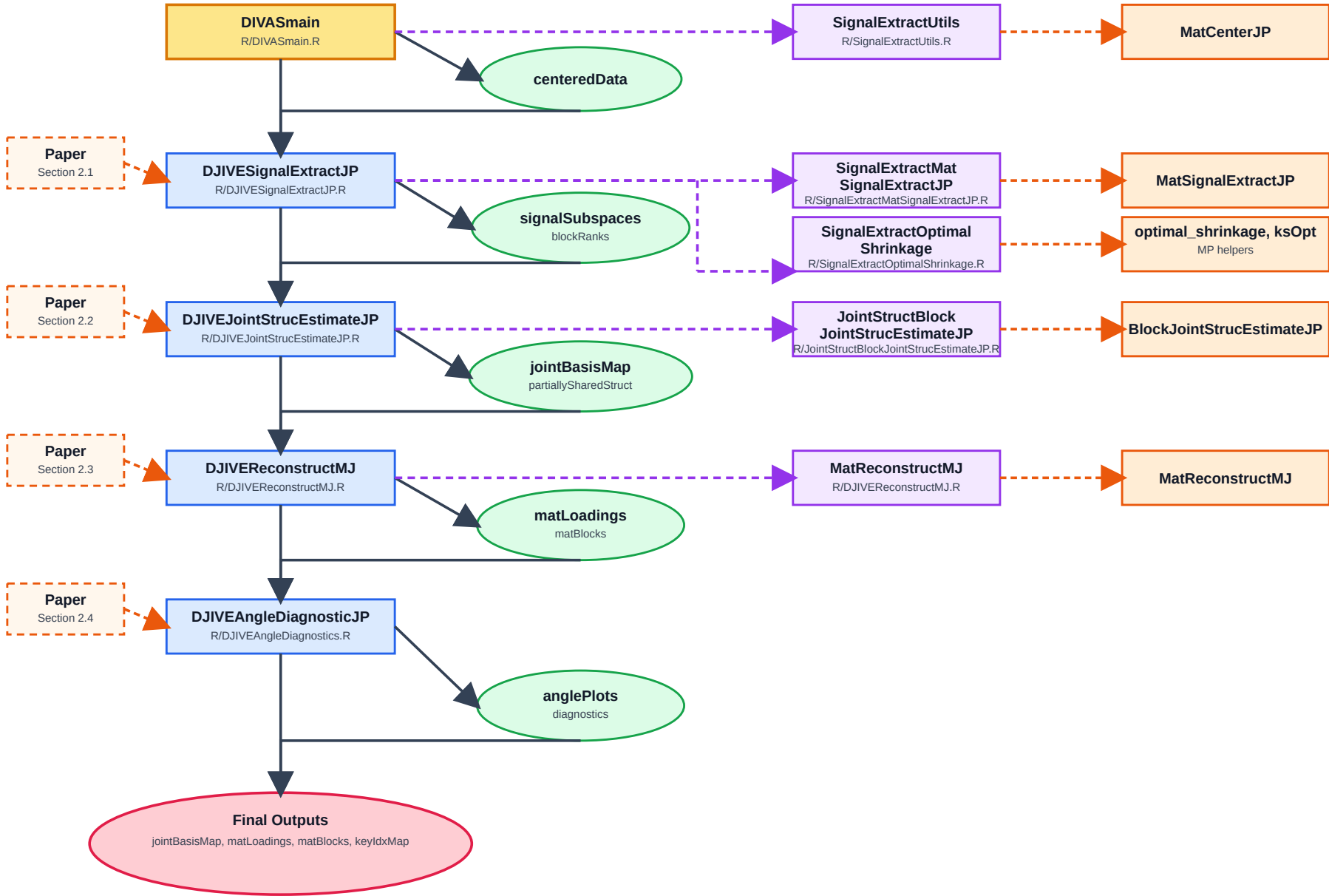
Paper Sections

Main Functions

Outputs

Code Files

Helpers



Paper Section Map

Paper Section	Main Idea	Primary Functions / Files	Helpers Used
Section 2	Overall DIVAS workflow and top-level pipeline	DIVASmain; R/DIVASmain.R	MatCenterJP; DJIVESignalExtractJP; DJIVEJointStrucEstimateJP; DJIVEReconstructMJ
Section 2.1	Signal subspace extraction for each block	DJIVESignalExtractJP; MatSignalExtractJP; R/DJIVESignalExtractJP.R; R/SignalExtractMatSignalExtractJP.R	optimal_shrinkage; ksOpt; randDirAngleMJ; MatCenterJP
Section 2.1.1	SVD, shrinkage, noise estimation, and rank estimation	MatSignalExtractJP; R/SignalExtractMatSignalExtractJP.R	optimal_shrinkage; MedianMarcenkoPastur; PercentileMarcenkoPastur; ksOpt
Sections 2.1.2–2.1.3	Perturbation angles and rotational bootstrap	MatSignalExtractJP; R/SignalExtractMatSignalExtractJP.R	randDirAngleMJ; bootstrap angle caches; residual imputation logic
Section 2.2	Joint and partially shared structure estimation	DJIVEJointStrucEstimateJP; R/DJIVEJointStrucEstimateJP.R	BlockJointStrucEstimateJP; penaltyCCPJ-PEarlyStop
Section 2.2 (optimization)	Sequential search and constrained optimization	BlockJointStrucEstimateJP; penaltyCCPJ-PEarlyStop; ccpSubOptJPEarlyStop	CVXR solver calls; block-subset search logic
Section 2.3	Reconstruction of component matrices and loadings	DJIVEReconstructMJ; MatReconstructMJ; R/DJIVEReconstructMJ.R	Re-rotation via svds; least-squares loading recovery
Section 2.4	Diagnostic displays and interpretation	DJIVEAngleDiagnosticJP; R/DJIVEAngleDiagnostics.R	randDirAngleMJ; rank image construction; ggplot output
Appendix A	Marchenko-Pastur support for shrinkage and noise estimation	R/SignalExtractOptimalShrinkage.R	incMarPas; MedianMarcenkoPastur; PercentileMarcenkoPastur
Appendix D	Optimization problem details	R/JointStructBlockJointStrucEstimateJP.R; R/JointStructPenaltyCCPJEarlyStop.R	ccpSubOptJPEarlyStop; CVXR::psolve

File Inventory

File	Role in Pipeline	Used By
R/DIVASmain.R	Top-level orchestrator	Entry point for full analysis
R/DJIVESignalExtractJP.R	Multi-block wrapper for Step 1	Called by DIVASmain
R/SignalExtractMatSignalExtractJP.R	Per-block signal extraction	Called by DJIVESignalExtractJP
R/SignalExtractOptimalShrinkage.R	Shrinkage and MP helpers	Called during Step 1
R/SignalExtractUtils.R	Centering and small utilities	Called in preprocessing and bootstrap setup
R/DJIVEJointStrucEstimateJP.R	Multi-block Step 2 driver	Called by DIVASmain
R/JointStructBlockJointStrucEstimateJP.R	One-subset joint search logic	Called by DJIVEJointStrucEstimateJP
R/JointStructPenaltyCCPJEarlyStop.R	CCP / penalty optimization routines	Called inside Step 2 search
R/DJIVEReconstructMJ.R	Reconstruction of outputs	Called by DIVASmain
R/DJIVEAngleDiagnostics.R	Plotting and angle diagnostics	Called after model fit
R/getTopFeatures.R	Downstream feature ranking helper	Used after final loadings